# Group 10: Laboratory 3

## Frequency domain filtering

Keegan Rolls (202004149)
Nicholas Philpott (202018248)

*ECE 7410: Image Processing*

*Memorial University of Newfoundland*

July 15th, 2025

## Low pass and high pass filter implementation

1. *Download the test image (puppy.jpg) and lp hp filters.m from Brightspace under Lab 03. Look at the image, and read the code and understand the implementation.*

Figure 1: `puppy.jpg`

2. *Read the image, convert the image to grayscale, and display.*
3. *Obtain the padding parameters. Display them.*
4. *Obtain and display the FT of the image.*
5. *Design the filter.*
6. *Implement the following filters (a. to f.): multiply the FT of the image with filter, undo the padding, move the origin of frequency spectrum to the centre and display the results.*
   a. ***Ideal*** *low pass filters for cut-off frequencies (0.3 and 0.7 of image height).*
   b. ***Butterworth*** *low pass filters for cut-off frequencies (0.1 and 0.5 of image height) and order n = 1, 5, 20.*
   c. ***Gaussian*** *low pass filters for cut-off frequencies (0.1, 0.3 and 0.7 of image height).*
   d. ***Ideal*** *high-pass filters for cut-off frequencies (0.1, 0.3 and 0.7 of image height).*
   e. ***Butterworth*** *high pass filters for cut-off frequencies (0.1 and 0.5 of image height) and order n = 1, 5, 20.*
   f. ***Gaussian*** *high pass filters for cut-off frequencies (0.1, 0.3 and 0.7 of image height).*
7. *What can you observe when increasing the cut off frequency radius of the* ***Ideal*** *filter?*
8. *What can you observe when increasing the order of the* ***Butterworth*** *filter when the cut-off frequency remains the same?*
9. *Discuss the general performance of the* ***Gaussian*** *filters in comparison to the* ***Ideal*** *and* ***Butterworth*** *filters. You do not need to discuss each possible comparison case. Only discuss any interesting phenomenon or performance observations.*